# Mingyu Kim (김민규)

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## Research Interest

My research interests include Vision-Language-Action models for robot manipulation, EEG-based emotion recognition and brain-computer interfaces, and human-robot interaction — with a long-term goal of working at the intersection of affective computing and embodied physical AI.

## Education

**Inha University** Republic of Korea
*B.S. in Artificial Intelligence Engineering (in progress)* *2024 – Present*

## Experience

**Inha University** Republic of Korea
*Undergraduate Researcher — Vision-Language-Action Robot Manipulation* *2025 – 2026*

*OpenVLA-OFT fine-tuning and deployment on a UR5e arm · with Changwoo Kim · Advised by Prof. Gihyun Joo*

- Built the full data-to-deployment pipeline for real-robot pick-and-place: ROS2 bag recording of RGB, joint states, and gripper I/O → conversion to RLDS format via a custom TFDS dataset builder → LoRA fine-tuning of OpenVLA-OFT with an L1 regression action head and proprioceptive input → real-time inference over a ROS2/ZeroMQ four-process architecture.
- Diagnosed and resolved critical training–inference mismatches — an unloaded proprioceptive projector, mismatched task-label formatting, an overly strict joint deadband suppressing valid motion commands, and an action-chunk indexing bug — that were causing the model to reproduce the current joint state instead of predicting motion.

**Inha University** Republic of Korea

*Undergraduate Researcher — EEG-Based Emotion Recognition* *2026 – Present*

- Implemented and systematically compared multiple baseline deep learning architectures for EEG-based emotion recognition using the open-source LibEER framework, evaluating their relative behavior and characteristics for valence/arousal classification.
- Building this comparative baseline foundation toward future work on affective, intent-aware brain-computer interfaces.

## Skills

Languages & ML: Python, PyTorch, TensorFlow/TFDS, scikit-learn, C++

Robotics: ROS2, real-time sensor/actuator pipelines, UR5e manipulation

Tools: Git/GitHub, Linux, LoRA fine-tuning, RLDS/TFDS dataset engineering, LibEER